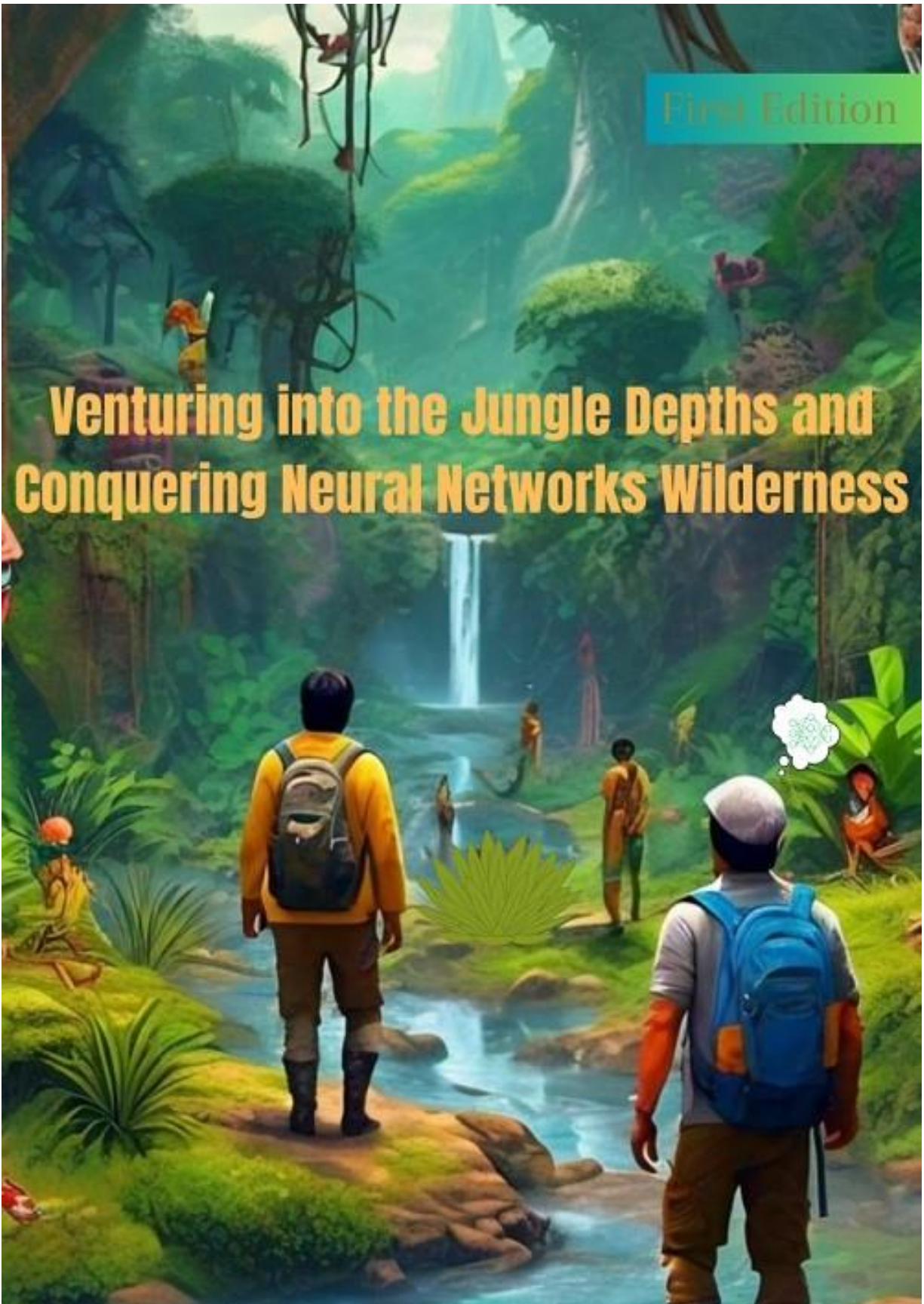


First Edition

Venturing into the Jungle Depths and Conquering Neural Networks Wilderness



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PREFACE

"Venturing into the Jungle Depths and Conquering Neural Networks Wilderness" is a journey, where we aim to decipher various concepts of Artificial Neural Networks (ANNs), especially Deep Neural Networks. In this adventure, we probe into Multi-Layer Perceptrons (MLP), Convolutional Neural Networks (CNNs), Autoencoders, Generative Adversarial Networks (GANs), and Transformers.

Narrating a fictitious tale of the journey of two travelers who explore different tribes within a mysterious jungle. Each tribe represents a distinct concept of neural networks. The characters and names used in this book are entirely fictional. However, the concepts discussed are grounded in the context of Artificial Intelligence. This creative approach aims to provide a unique and interactive learning of the complexities of Neural Networks, making it accessible to everyone curious to explore AI and its applications.

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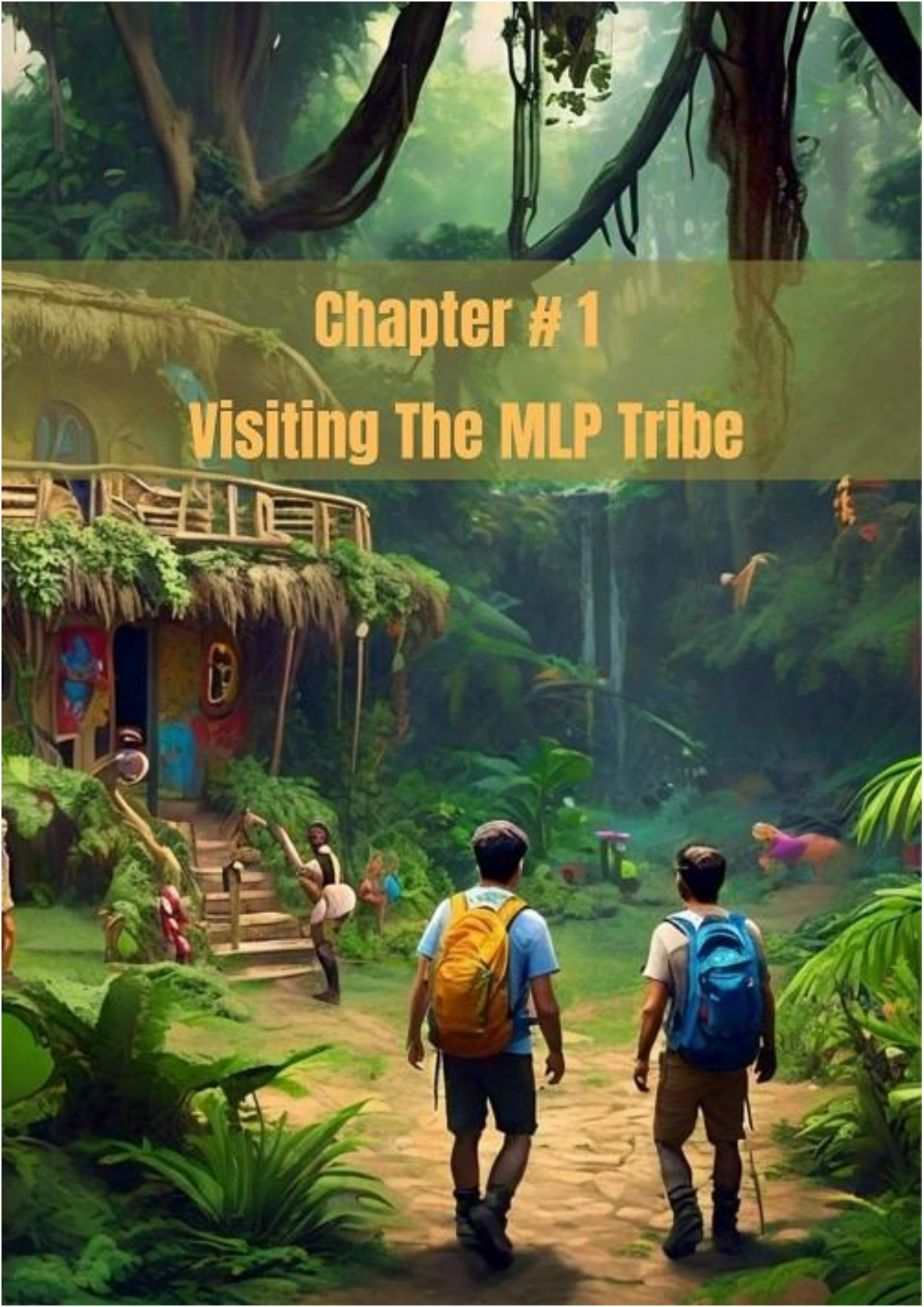
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A lush, vibrant jungle scene. In the foreground, two characters with backpacks (one yellow, one blue) walk away from the viewer along a dirt path. To the left is a small, colorful hut with a thatched roof and a balcony. Several indigenous people are visible near the hut. The background is filled with dense tropical foliage, a waterfall, and a large bird in flight. The scene is brightly lit with sunlight filtering through the trees.

Chapter # 1

Visiting The MLP Tribe

IN THE UNKNOWN: JOURNEY BEGINS

A gentle sea breeze rustled the neem tree's branches, while the wind whispered softly outside the window. Talha, a young university student, was engrossed in his studies, when the soft twinkling of wind chimes suddenly startled him. His face showed signs of tiredness, and his eyes were serious. He was completely worn out from the pressure of his tough academic routine, which confined him to his room for studying and prevented him from engaging in any physical activity. Frustration has been a faithful companion till late, but he longed for change and motivation that could boost his energies and reverse his deteriorating academic progress.

Talha was enticed by the soft tinkling of windchimes and approached the window to enjoy the breeze. As he did so, he noticed that his pain seemed to be lessening and pondered how spending time in nature may have a positive effect on one's emotions. As he considered this, he thought back to what his old friend had said about the transformational force of nature. All of a sudden, he made the decision to put his books down, step outside and enjoy nature.

Talha spent many hours wandering the city in quest of comfort before ending up for a stroll at a park. While out for a stroll, he noticed a booth with a banner showing several routes leading into the forest and wilderness, and another banner urging people for exciting jungle adventures. His worn-out eyes opened with delight and newfound hope as he saw hidden wonders, enigmatic forests, and twisting trails. Curiosity and determination drove him to seek out the woman in charge of the booth in order to learn more. Half an hour later, armed with the knowledge gained from the woman at the booth, including the information about the trek guide Moiz, and a map of exotic places, Talha wandered through the neighborhoods of the central city near the municipal library searching for Moiz in teashops. Eventually, Talha found

Moiz in a teashop tucked away in a peaceful area close to the municipal library.

Inside, the teashop had a serene atmosphere, and Talha was greeted by Moiz himself, a man with a wise and comforting smile. Talha was delighted to find Moiz as amiable as described by the booth attendant, a friendly and experienced guide of wilderness and nature. The warm welcome encouraged Talha to eagerly request Moiz for sharing his adventurous experiences. Talha listened to Moiz's personal stories of life-changing encounters in the middle of the wilderness, while enjoying the serene setting of the teashop. Talha's excitement was clearly notable. On the other hand, Moiz's eyes gleamed with the knowledge of someone, who has a deep relationship with nature. Talha was inspired by Moiz's tales and his love for nature, and he sought Moiz's willingness to accompany him on his upcoming jungle adventure.



For Talha, Moiz seemed to hold the key to unlocking the mystery of his wornout soul. Sensing Talha's excitement and yearning for change, Moiz agreed to join him on his expedition. They discussed and finalized the route,

expedition details, meeting place, and time. As Talha left the teashop, he felt a surge of joy and happiness, a stark contrast to the dejected and gloomy person he had been for the past several days.

The next day at dawn, they met at the city exit and began their journey, following the route marked on the map. At the edge of a dense forest, shrouded in shadows and alive with the chirping of crickets, Talha and Moiz took a deep breath and embarked into the forest's depths. The path unfolded before them like a hidden trail, waiting to be discovered. As dusk fell, they set up camp in a clear and peaceful area, following Moiz's guidance and the map. The tranquility was broken by the gentle crackle of the campfire. For Talha, the fulfilment that had been eroding under the rigid confines of academics began to blossom like a resilient flower in the outdoors. Under the starry sky, they enjoyed the forest's stillness and eventually drifted off to sleep.

The Jungle Map



Encountering the MLP Tribe:

Next morning, they advanced deeper into the heart of the dense forest, enveloped by the sounds of the jungle as the air grew heavier with each step. Towering trees filtered the sunshine with their gnarled branches, engulfing them within the shadowy folds of the trees. They came to a tiny, open space surrounded by odd flora and vibrant flowers, and a little distance away was the home of an unidentified tribe. They noticed an odd shift in the atmosphere as they got closer to the tribe almost as if the forest's spirit had congregated in this unique and enigmatic location. From the tribe's entryway, they could observe individuals going about their daily business.

The people's clothing appears to be made of plant stuff. While women were gathering some herbs with care and expertise, men in another area were preparing mixes using the herbs the women had collected from large, antique pots. The tribe's members gradually began to notice Moiz and Talha and grew wary. One of the men began to approach them cautiously. Despite his advanced age, he possessed a strong physique and solid gait. His face was scarred by time, yet knowledge glowed in his eyes. He smiled in acknowledgment and asked them what their goal was. Moiz then informed him about their journey.



He greeted them warmly and dubbed his group "The MLP^[1]." Moiz and Talha saw how the tribe members interacted with the surrounding environment as they strolled through the community. They were interested by this, especially Talha, who saw connections between the natural world and the artificial cerebral constructions he had studied in college. With great enthusiasm, Talha asked the tribe's leader, the older tribesman, to show them their special method of living in the bush. In answer to his request, they received guidance on how to identify various plant characteristics and comprehend their therapeutic qualities. Given that the jungle is a type of natural habitat made up of several strata that combine to form an ecosystem, it is home to a variety of living things, including humans. Each layer is necessary for the other layers as well as for the life of organisms. Every organism learns from its interactions with other organisms how to cohabit in the ecosystem in order to thrive. Depending on the species involved, the nature of these interactions may change. Furthermore, different ecosystem strata have different species counts and types. An artificial neural network^[2] with several layers and a variable number of nodes or perceptrons^[3] in each can be compared to this environment.

Nature's Whisper: Feature Extraction

After discussing the tribe's history of expertise and how they came to be known as MLPs the chief of the tribe focused on the women's aptitude at recognizing and distinguishing various medicinal plants from one another. These women collected each herb plant separately by using their acute senses to identify differences between them. Talha saw right away that the woman's expertise and the functioning of an artificial neural network's perceptron were similar. The basic role of multi-layer perceptrons often called pattern recognizers and commonly used for classification^[4] and regression^[5] in machine learning, is to distinguish patterns within an image.

One must comprehend the system in order to develop various herbs that will flourish in this vigorous ecosystem. A portion of this system's layers remain unseen to humans, but they are essential to the upkeep of these herbs since they help turn a particular herb plant's seeds (input) into healthy plants (output). The deep soil layers that are inaccessible to people need sunlight, water, and even contact with other animals to support their growth and general well-being. However, one must take into account the degree to which these interactions are necessary. Are all of these necessary, or are they only necessary in particular proportions? It is necessary to determine each of these elements for a successful crop.

The chief of the tribe went on to describe how the tribe had accumulated knowledge over many generations about how much sunlight, water, and other resources are exactly right for each plant to thrive healthily. Talha was occupied with relating this information to the function of node weights and biases in the MLP layers, particularly in relation to how they link distinct nodes in different MLP layers. He recalled that activation functions are used to learn these biases and weights within the hidden layers of MLP. He also mentioned a few MLP neural network activation functions^[6], such as sigmoid, ReLu, and tanh. According to Talha, the tribe members serve as a catalyst for the growth of herbs. "Forward propagation" refers to the entire process of planting a plant's seed in deep soil, keeping the balance of light,

water, and other essentials by the tribe, and allowing the soil's layers to interact with the seed to support its growth into a full plant. In this process, the seed acts as the soil's input and yields a plant as its output.

The younger members of the tribe make use of the calculated amounts of sunlight and water needed for particular herb plants. If the harvest is poor as a result, these people reevaluate the proportions they have chosen and adjust by becoming aware of the ecosystem's changing needs as time passes. They then put their newly acquired information to use by controlling the amounts of water and light while attempting to reduce harvest losses. It may take several iterations of this process of ecosystem adaptation before a successful harvest is achieved. This function is similar to that of back propagation^[7] in the MLP system in that it seeks to minimize error by computing the error gradients with respect to the network's biases and weights and updating them in a way that minimizes the error of the network. Additionally, a group of knowledgeable tribesmen regularized this procedure such that the cultivation land could be utilized for multiple harvests, provided that the amounts of light and water did not adversely influence the soil's health.

Talha was thrilled to be able to connect his academic understanding to the practical scenario. Talha and Moiz ate dinner with tribe members in a cool, welcoming setting after a hard day. After everyone had gone to bed, Talha grabbed his paper and jotted down all the things he had learned that day, along with an illustration of MLP's^[8] block diagram (Figure 1). He was already looking forward to the next day, when he and Moiz would follow the map's instructions and explore more of the woodland.

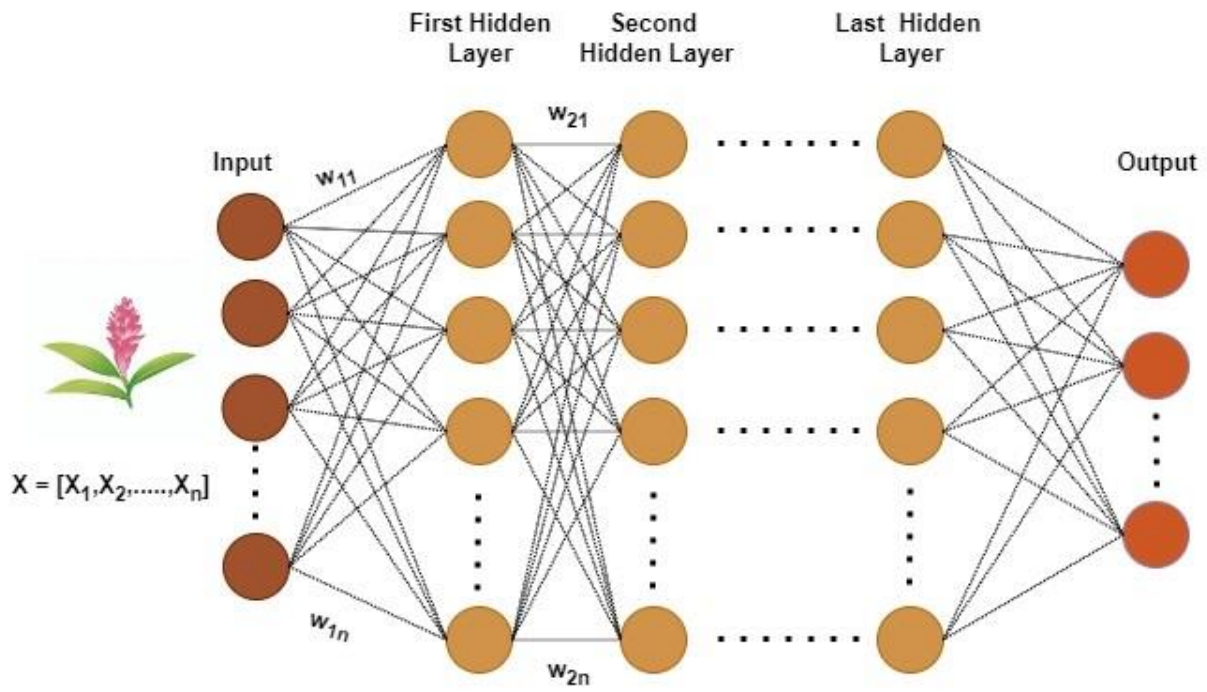
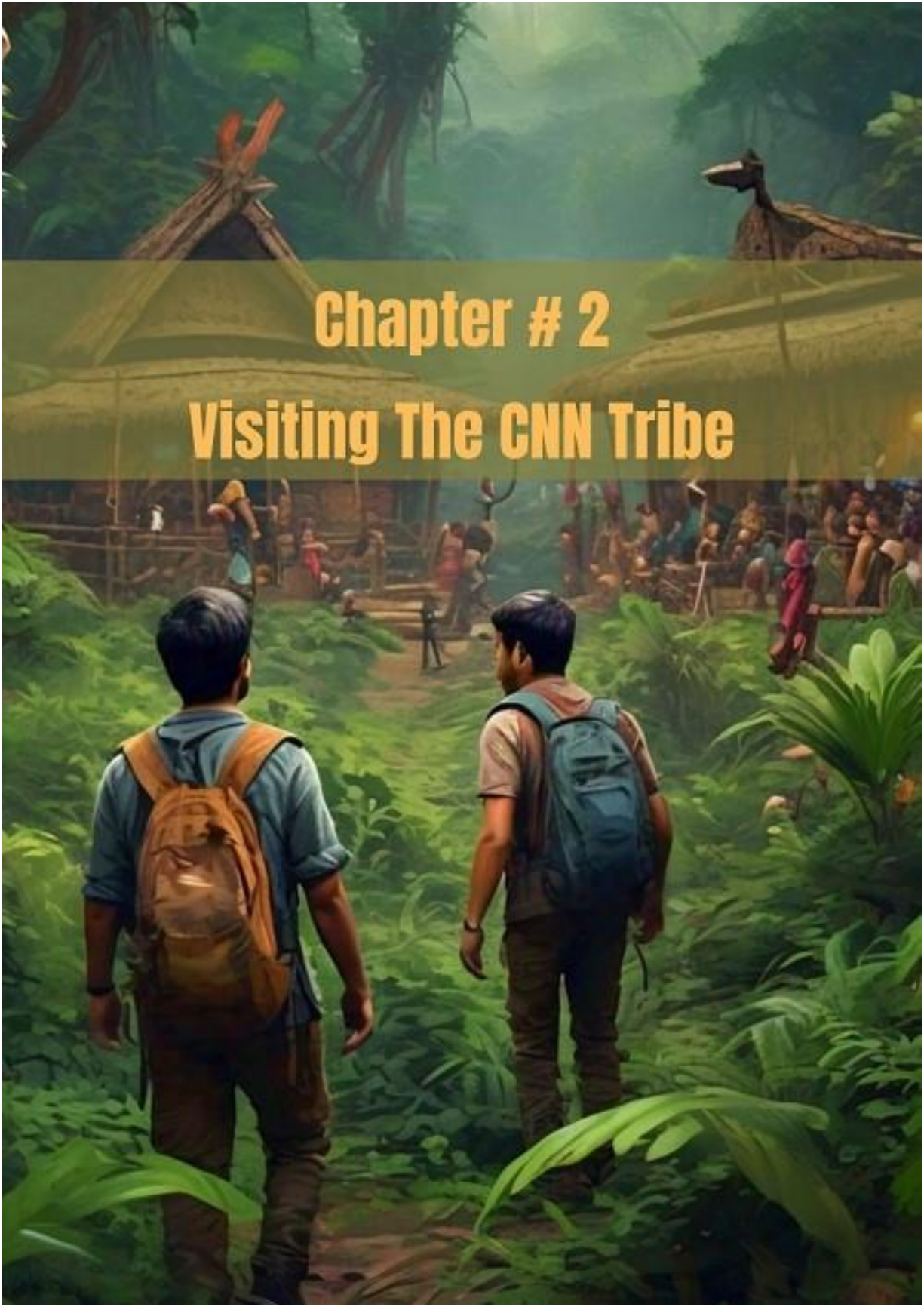


Figure 1: Block Diagram of Multi-Layer Perceptron

Narrative Arc:

Eager to unravel the mysteries of the deep jungle, two daring explorers, Talha and Moiz, step into it. When they come across the MLP tribe, they learn about the fundamentals of neural networks. They discover the basic components of artificial intelligence, including neurons, activation functions, and the interwoven layers that make up the MLP tribe, as they explore the working mechanism of the MLP tribe.

The background image is a vibrant, painterly illustration of a tropical jungle. In the foreground, two men are seen from behind, walking along a dirt path. The man on the left wears a light blue shirt and brown pants, carrying a large brown backpack. The man on the right wears a tan shirt and dark pants, carrying a blue backpack. They are walking towards a village in the distance, where several thatched-roof huts are visible. The jungle is filled with dense green foliage, including large ferns and trees with thick canopies. The lighting is soft, suggesting a misty or early morning atmosphere. The title text is overlaid on a semi-transparent yellow-green band across the middle of the image.

Chapter # 2

Visiting The CNN Tribe

JOURNEY RESUME'S TO EXPLORE NEW TRIBES:

As the sun rose, Talha and Moiz went deeper into the forest with sounds of birds and insects surrounding them. Eventually, they took a rest to drink water from the stream. While sitting on the rock, they exchanged their acquired facts learned from the MLP tribe. Talha was happy to be able to relate the workings of MLP tribe to the basic concepts of a neural network that he had studied at the university: concepts such as activation functions, hidden layers, weights, neurons (the building blocks), inputs, and outputs of the first basic model or architecture - the Multi-Layer Perceptron.

They resumed their journey, and as they walked, time passed slowly until the sun shone brightly on the jungle. Both started to feel weary under the bright sun. As they were about to take another rest, they found themselves before a twisted pathway covered with bushes full of thorns. It looked difficult to cross. After some deliberation, they walked along the path very carefully avoiding the thorns. Eventually, they crossed it but what lay ahead assured them that they had reached their next destination: the new tribe of their neural network.

As they moved further into the tribe, they observed that the place was very quiet. They continued moving and observed their surroundings. They saw that everything there was well-structured and separated by some kind of categorization. At some distance, they saw a group of people sitting leisurely and enjoying a cup of tea. On approaching them, they introduced themselves and politely requested to meet their leader. One of them kindly offered to lead them to their leader.



On their way to the leader, both Talha and Moiz continued to observe the people around them. What caught Talha's eye, was the way the people were dressed. Their clothes looked like they were crafted from a very special type of leaves, each of a different color. He surmised that these colors may hold kind of implication. Talha was very excited but felt a confused by his surroundings.

When they met their leader, they explained that they were travelers who had explored the MLP tribe and had now reached here. The leader was very cordial inquired about their intentions. Talha requested to see his tribe be allowed to observe their way of living and their daily life tasks routine. The leader happily accepted his request and invited them to their tribe, which they named as the "CNN" tribe. The name triggered Talha's mind, and he thought about Convolutional Neural Networks (CNN)^[9], which he had studied at university.

The leader first entertained them with food and tea. After they had enjoyed their tea, the leader started the tour of his tribe while explaining the history

of his tribe. He explained how within the tribe, everything is grouped and divided into categories based on their qualities and similarities. He also specified their connection with the MLP tribe and how they assist in resolving problems that cannot be solved by the MLP tribe themselves. Talha and Moiz keenly listened their story.

Unearthing the Essence: Feature Extraction[10]

As the leader continued their tribe's story, he led them to a vast place where a large number of people were working in groups. The leader explained that every group had specific tasks to do. In the first group, the 'Extractors', people observed and noted key characteristics of the items they processed. Talha saw a pair of women meticulously examining a fruit. The woman's thorough examination appeared like sliding a visual filter over the image by concentrating on a portion of the image at an instant. And then sliding this window of focus on other patches of image, extracting only relevant information. This trait inspired the name "Convolution" for the tribe, analogous to CNN's convolution operation. Talha was told that it is the collective experience that determines the spatial extent (size) of the convolution. In the native tribe's language, this unique sliding filter was also referred to as a "Kernel".

After a thorough check, the women recorded the fruit's properties and passed it to another pair, along with their observations. The next pair added more observations to the previous ones by inspecting the fruit and then passed it to another pair for further analysis. These observations facilitated not only in the identification of different fruits but also in perfecting their production by symbolizing specific features of each fruit.

Talha now understood how CNNs work: different layers extract various features using convolution and by sharing weights, local features are extracted from the data. The progression of the fruit from the first pair to the last, for examination and the extraction of unique features, parallels the function of layers in CNNs.

Seeing the Unseen: Pooling

They moved further to another small group. This group received all the symbols that were extracted from the Extractor group. They assembled these symbols into compact forms. This method helped to reduce complexity and the size of data passed onto to other groups for further processing. Their main function was to gather only those symbols that were more important for the identification of the item and discarded unimportant symbols. At this moment, Talha understood the concept of pooling in CNN, how pooling reduces the dimension of data which makes it more robust. He shared this concept with Moiz. Moiz further elaborated on the concept to Talha explaining that pooling also made the CNN network more robust to changes. He explained that if the same fruit were brought to the pooling^[11] group next time in a group of similar fruits, the pooling group would be able to recognize this fruit in the correct category. He gave Talha an example of similar fruits like apples and pears. Talha was very happy about how learning improved at every step, and it made him more excited to match the knowledge he gained with what he learned at the university. He became excited when the leader told them that this group also have a unique name called "lufwaya amanos", which means "Seeing the unseen".

Aligned Pathway: Concept of Flattened Data

Now the leader took them to a new group. Talha and Moiz observed that this group was not as big as the one they saw before. The leader told them that this was like a delivery point. Here, symbols extracted from previous groups were assembled in a special format so they could be delivered to the next group. At this, both Talha and Moiz were surprised and looked at each other. They spoke at unison, mentioning “flattened”, another concept of CNN, in which the extracted data were flattened to make it a feasible input for the next step. They laughed, and the leader was very cordial and inquired about their intentions. Talha then asked about the name of this group. The leader told them the name as “Grid Guidance: Mapping the Aligned Pathway”.

Now the leader told them to take rest for some time, and then they can continue their journey to the final group of this tribe. The leader offered juices and fruit to them. They rested for almost two hours. As the time passed, darkness started spreading, but they both wanted to explore this tribe before night. They requested the leader to resume their tour of the tribe. By seeing the excitement of both Talha and Moiz, the leader took them to the next group without any delay.

Navigating the depth: Fully connected layer

The leader told them that data from Grid Guidance, where important information from previous groups was gathered, flattened and organized and efficiently passed to this group. Each data had its own significance. People in this group carefully examined each data point to ensure no detail was missed, as it aided in making the correct decision, crucial for the tribe's life. They continued to inspect and update the data until they reached the most accurate decision. This process was repeated until no further changes occurred ensuring they arrived at the correct decision. This activity was similar to the concept of the fully connected layer, that Talha could recalled but he could not fully grasp the understanding.

Moiz assisted him by providing elaboration. He described that when fully connected layers^[12] received data, they aimed to classify an object like a fruit, such as an orange. They examined data until the orange is correctly classified. This involved checking different attributes like structure, color, etc., with the help of the given data that they already had. These attributes can be referred to as weights, biases and activation functions which helped them to understand the complex structure of data. And by updating the weights with the help of an activation function like sigmoid or Relu, loss of incorrect classification was reduced, ultimately leading to the desired results. Moiz further explained that the data points were like the neurons in the fully connected layer.

Adapting to new Habitat: Transfer learning

In the end, the leader led them to another group that helped save time for similar tasks. This group comprised of the most experienced people who did not require the entire process for classification of object seen in the groups observed previously. For objects within the expertise of this group, they did not have to pass through all the previous groups; instead, they were directly delivered to this group for classification. Nevertheless, if unfamiliar object was discovered then the expertise of every group was needed. Talha appeared a bit confused but Moiz helped him understand by simplifying the concept comparable to transfer learning, in which pretrained model was utilized without the need of any new data to classify objects. We employed existing architecture and fine-tuned weights to get desired results.

With a sense of accomplishment, they realized that each discovery they made had brought them closer to unraveling the mysteries of the neural network. After an exciting tour and a bit of tiredness, with night spread all around, Talha and Moiz joined the tribe gathering, and had dinner in a very peaceful environment gossiping with tribe people.

After the tribe members went to sleep, Talha cataloged the day's journey and new perceptions. He depicted the whole mechanism with a diagram as shown in Figure 2. As Talha and Moiz lay down to rest, they thought about the new journey ahead, their minds filled with excitement for what tomorrow might bring. The next morning, the sun rose high in the sky, signaling a fresh day of adventure. Talha and Moiz walked away. They set off again, their joy and newfound understanding propelling them onward. They were curious to learn more about other tribes and their future. They couldn't help but be enthusiastic about the wonders that awaited them as they ventured into the unknown. A glow of excitement filled their faces as they began the next leg of their journey.

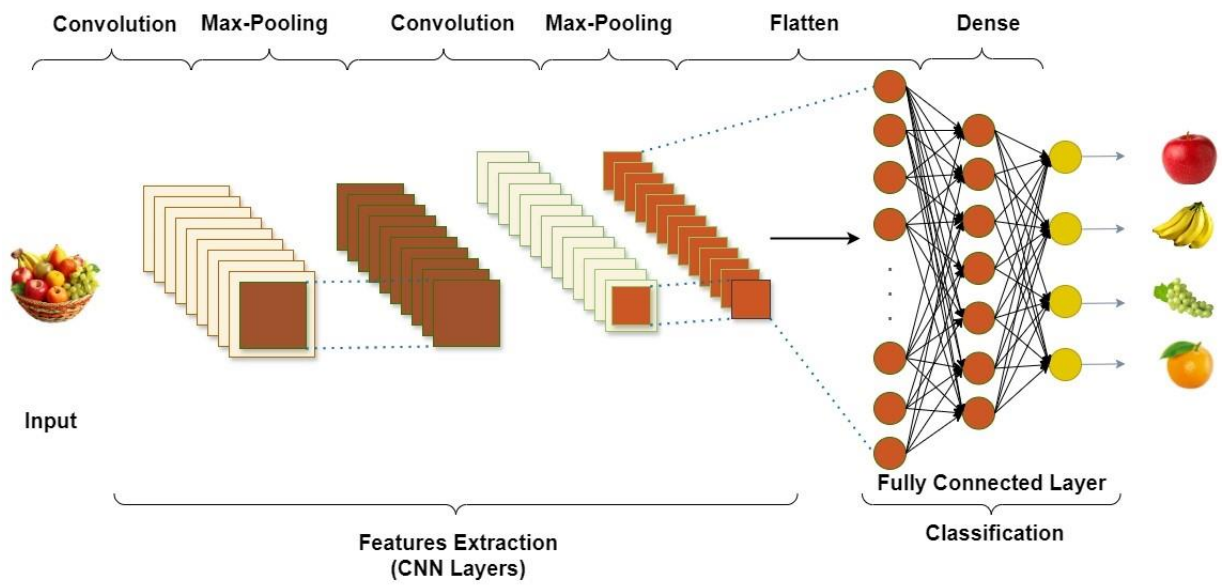
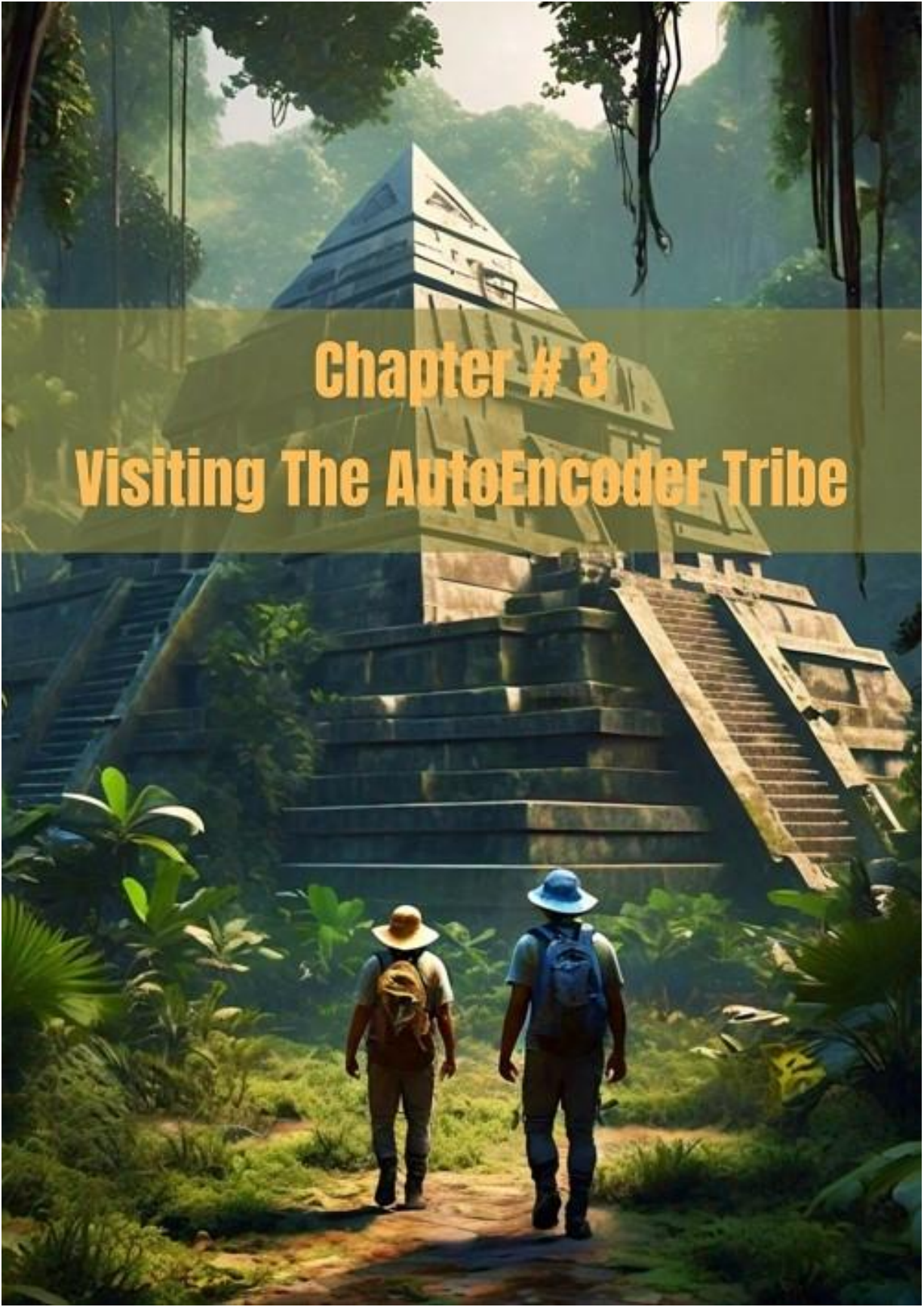


Figure 2: Block Diagram of Convolutional Neural Network

Narrative Arc:

As they continue to explore the forest, Talha and Moiz come across the CNN tribe and become completely engrossed in the world of convolutional neural networks. Here, they see how the CNN tribe can identify patterns in photos and data, showcasing the potential of feature extraction and categorization.



Chapter # 3

Visiting The AutoEncoder Tribe

A MYSTERIOUS TRIBE

Talha and Moiz walked for a couple of hours, and as they grew tired, they sat under the shade of trees and ate snacks that tribe members had given them for their journey. After a quick nap, they continued their journey towards the next tribe, following the map directions. As night approached, both of them became worried about where to stay if they couldn't reach their destination in time. They were in a dense forest surrounded by the sounds of wild animals. They were weary, and their clothes were sticking to their skin due to humidity. The air was still, signifying that rain could start at any moment, adding to their misery. Eventually, it began to rain, and they hastened their pace while searching for shelter. In time, they spotted a cave. Feeling relieved, they dashed towards it, entered, and set down their bags. As they prepared to settle in, they heard noises resembling people talking and strange festival music.



At first, they were frightened, but then they decided to check what was going on. They moved to the other side of the cave and were surprised; their eyes widened because they saw a completely new place. People were gathered around a fire, sitting on either rocks or a plain surface. They seemed like the elders, respected individuals of authority.

They approached those people and were warmly greeted by them. The elders readily offered them shelter for the night, as well as water and fruits to quench their thirst and hunger. They explained to the elders that they were travelers on an adventure to explore different tribes related to Neural Networks. They shared their journey from previous tribes and how they had

arrived here. While they were enjoying their meal, they overheard the conversation of those people discussing some kind of data and features.

Those people were both enthusiastic and passionate about what they were discussing; it seemed like a very dear topic to their hearts. Upon inquiry by Moiz, one of the most authoritative-looking men told them to rest for the night, assuring them that all their inquiries would be answered in the morning and they would be given a tour of the tribe. He then instructed one of the men to guide them to their resting place.

Talha and Moiz woke up at dawn to the sound of birds chirping. They had breakfast with the tribe members, surrounded by the beauty of a bright sunny day with a slight breeze, greenery all around, and the fragrance of soil after the rain adding to the beauty of their surroundings. Nature was at its best. They were mesmerized by the beauty of the tribe.

As promised, around 8:00 am, the authoritative man, now confirmed by the local people to be the leader of the tribe, appeared and requested them to follow him. And so, the adventure began. The leader took them to different places of the tribe. They visited many areas. Everywhere they went, they observed that the environment was very clean, with calmness prevailing and no disorder. They saw many strange and unique patterns of woods and trees as well as rare and distinct birds and animals.

As they walked through the tribe, the wise man took them to their fields where they harvested, which was their uniqueness. Talha and Moiz found themselves mesmerized and surprised by the beauty of the cultivation, gently swaying with the breeze. Upon closer inspection, Moiz was astonished to see herbs growing in the fields that were almost entirely distinct from those in their conventional world. Moiz spontaneously asked the leader about their cultivation methods for the herbs widely used in medical treatments, mentioning their rarity in their conventional world and near-extinction. He gently but curiously probed the leader about their cultivation methods, expressing his fascination with how they preserved the quality of these herbs and achieved such precise cultivation. During Moiz inquisition, the leader was smiling with pride. He told them to be patient assured them that he would

share the tribe's methods once they reached the main hub of their activity. Proudly, he led them to that location. The place was enveloped by trees and branches adorned with small flowers, all interconnected like a network, with every node linked to another. The structure resembled pyramids, and in that moment, Talha and Moiz connected the bridge of their academic knowledge with these pyramid-like structures. They spoke in unison, 'Encoder Decoder!' The leader told them that this place was the heart of his tribe known as the “Grand AutoEncoder”. At the Grand AutoEncoder, people were working very carefully and with full concentration. There was hush silence and everyone was deeply engrossed in their work. Talha noticed that there were people of different age groups.

The leader took an “Epazote” stem, an herb from the collection of multiple herbs at nearby table and explain that he will show them how this Epazote properties will be identified and then be preserved for next cultivation.

Learning from the Past: Encoder

In the first stage, the youngest group of people keenly observed it, noting down its properties, such as whether it was in full bloom, its appearance (e.g., its size, approximately 1 to 2 centimeters), and whether it had a mild fragrance or not and passed this information to the next upper level for further analysis. The upper level comprised of a group of more experienced individuals belonging to a mature age group. The flower was reanalyzed by them and the findings were recorded, identifying additional features that the first group might have overlooked.

As the herb was analyzed at multiple levels and its features were logged, it reached a stage known as “latent space”. In this stage, the sole worker; the most experienced and skillful individual, diligently discarded recorded features that were not relevant to the perfect " Epazote " quality. He learned different characteristics of the herb and effectively encoded them into a compressed form. This compression mechanism was the focal point of the process, hence why the entire process is known as an 'Encoder'. Only the

most significant data was retained and passed to the next part of this Grand Auto Encoder.

Reconstruction of data: Decoder

The tribe's leader explained that the next hierarchy was followed a similar manner but in reverse; in the encoder, information passed from the lower level to higher level, but in this part, the compressed information flowed from the higher level to the lower level. He explained that the main working of this part was the reconstruction of data to its original they called this part "Decoder". The encoded characteristics of Epazote were decompressed at each level of decoder, and from these decoded characteristics, a premium seed of this herb was bred comprising all the desirable features.

The encoder-decoder can also be used to produce pesticides for different plants, categorize different qualities of a plant and determined soil composition required for specific plant. The wise man explained that by using this hierarchy or structure of working, they were able to preserve almost extinct plants and were able to grow these herbs in any season and in any weather condition. This was their mastery.

The Grand Auto Encoder and these people's effort inspired Talha and Moiz. They expressed their gratitude to the wise man for the journey and the wisdom he imparted. They requested permission from the wise man to remain for a few more days to learn more about the process of Grand Auto Encoder. The wise man accepted with joy, offering them a place to stay. n. Talha and Moiz decided to make the most of their time and learn as much as they could from this community, grateful for the opportunity.

After spending an exciting as well as tiring day, both Talha and Moiz lay down for rest. Talha opened his notepad and catalogued the working of AutoEncoder that he learned during the day from the tribe. He also illustrated a figure of an AutoEncoder^[13] (Figure 3) to gain a better understanding.

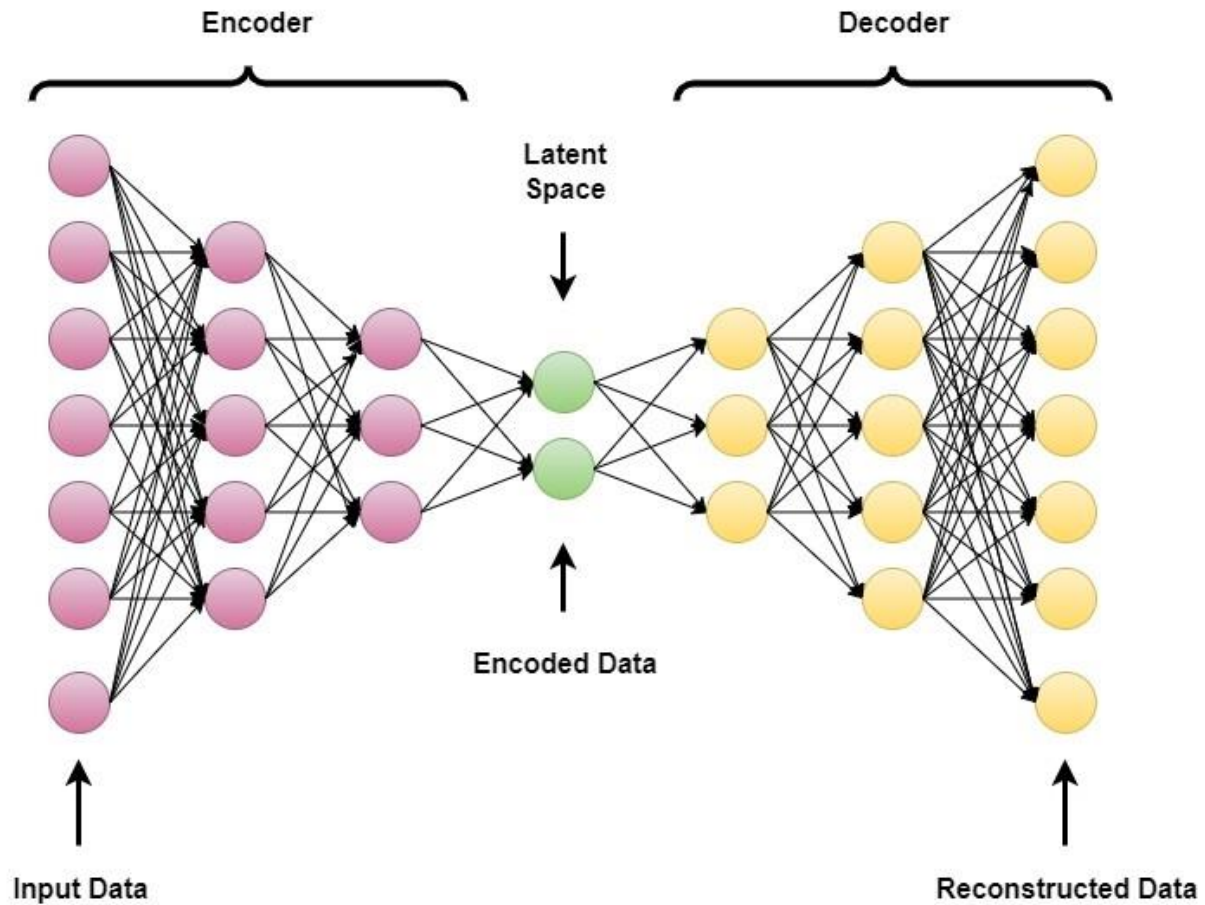


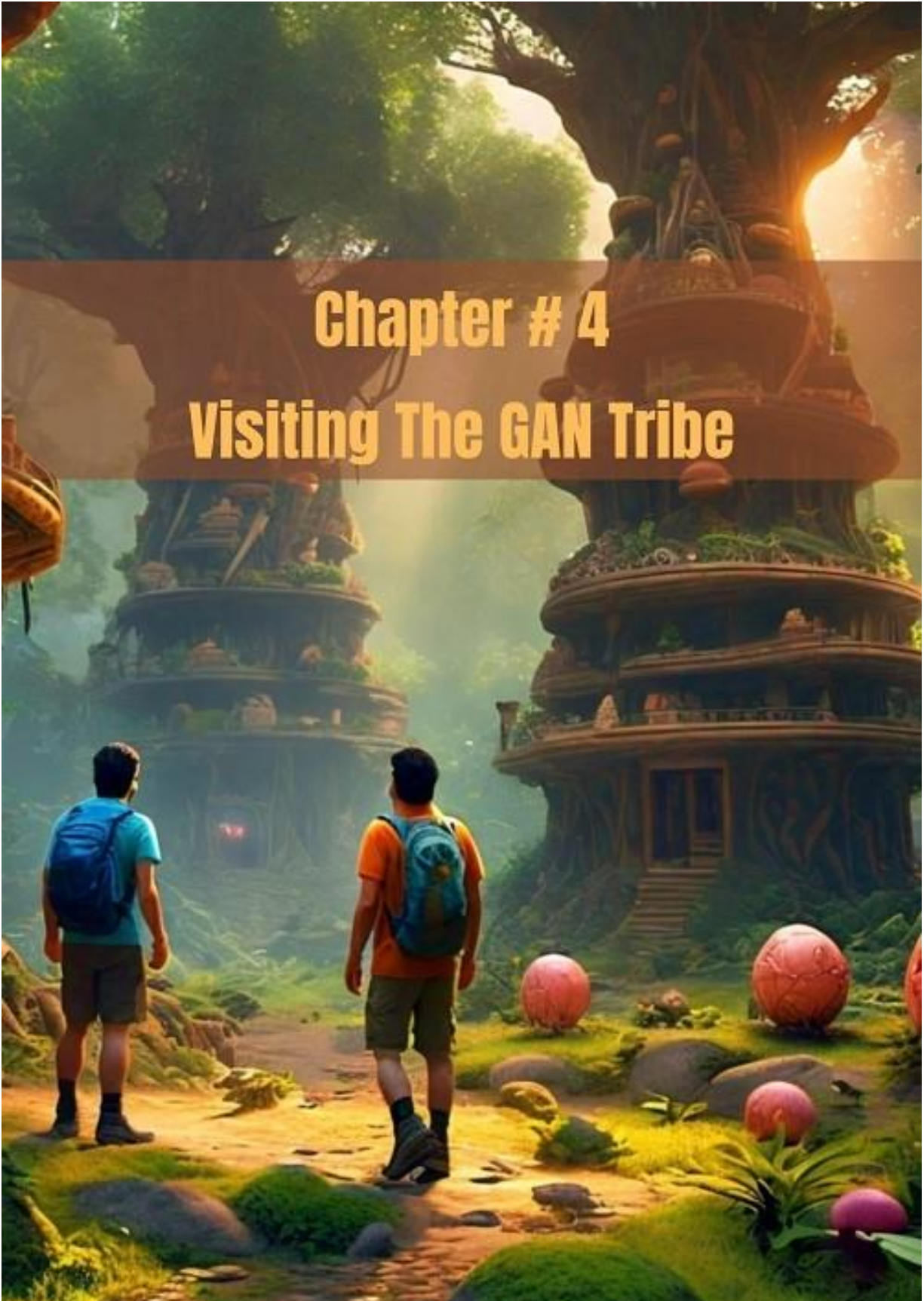
Figure 3: Block Diagram of AutoEncoder Architecture

Narrative Arc:

Talha and Moiz discover the Auto Encoder tribe's secrets as their journey proceeds. They learn how data can be compressed and reconstructed using the ingenious functions of encoders and decoders. This tribe teaches them about data representation, dimensionality reduction and the art of upholding essential information.

Chapter # 4

Visiting The GAN Tribe



ENCOUNTERING A MAGICAL TRIBE

Thankful for the bright sun and gentle breeze, Talha and Moiz walked through the dense forest, discussing the complex and laborious methods used by the auto encoder tribe. The map indicated that the next tribe would be found near the river's banks. Up until now, there was no evidence of a river. In the distance, dense trees surrounded them in all directions. About to take rest and searching for a spot to relax, they heard a rushing sound, like the flow of water. Excited, they hurriedly moved in the direction of the sound. Walking for some time, they saw what seemed like a whole new world. They saw a beautiful river flowing, its water sparkling. They checked the map and realized their next destination was just a few steps away from them. They washed their faces with the river water and walked along the bank of the river. After traveling for a few ups and downs on the bankside, they finally arrived at a tribe that held the promise of a new adventure—the GAN tribe. They saw some girls plucking flowers, and boys playing some kind of game. They walked casually in the tribe. They called a boy and asked him about their leader. The little kid took them to their wise man. The tribe leader was an old man who welcomed them with open arms and hospitality, inviting them to have dinner with him as dusk began to settle.

During the dinner, the leader told them about his tribe, the "GAN: Generative Adversarial Network."[\[14\]](#), and their way of life. His tribe members were experts in restoring old relics, mending broken scriptures, and reconstructing them in a way that made it difficult to distinguish between the original and the copy. Some experts could create multiple variations of an image. Moiz and Talha were captivated by the tribe's powerful trait that resembled the current machine learning model of GAN, which consisted of a generator and discriminator, both working adversarially against each other. Their capabilities lay in creating images that were identical to the originals but could also produce varied forms of those images. For example, if a picture of a person with white, blond, curly hair, a round face, and blue eyes was

input to the GAN, it was capable of generating a variety of images of the same person. These variations could include a person with brown, straight hair, brown eyes, and glasses; another with pale white skin, blue eyes, wavy hair, and a mustache; or even one with white skin, grey eyes, and a beard and mustache. All these variations looked like real images of different people, but in fact, they were fake and nonexistent.

Talha and Moiz eagerly requested him to show them his tribe's impressive and masterful working methods with excitement. The leader was quiet for some time, and they both looked at him with apprehension, fearing he might deny them. However, what he said next also dampened their spirits. He sadly explained that the real experts of GAN working were at odds with each other in real life, leading to a dilemma within the tribe. As a result, there had been no progress in their work for the past few weeks. After some contemplation, Moiz asked for details, and he told them that the two experts were brothers in real life and were not working together due to some minor disputes. The brothers, Saad and Hashir, were experts who worked like a generator and discriminator, respectively^[15]. Saad could replicate the original and its variants with perfection, while Hashir had the expert eye to identify imperfections, which in turn helped Saad to improve his skills.

Both strived for success in their work and to minimize their imperfections. The leader further elaborated that recently, they had discovered a broken scripture with some missing text. All the elders believed that this discovery was related to a holy custom that had been missing for years. Now, if the missing text could be generated and deciphered, it would enrich their knowledge of their customs and help them follow them as required. However, with this altercation between the brothers, everything was put on hold as they were key players who had skillfully led their team in the past. The elders had tried to bridge the gap between the brothers, but alas, all their efforts had been fruitless till now. The tribe leader even showed them the broken scripture. Moiz told the leader that they will try to avail the assistance of both and requested the leader to guide them to Saad and Hashir in the morning. The leader was skeptical but he agreed.



The next day, Moiz and Talha got up early and, after breakfast, went to the leader who was waiting for them with his young teenage son, Bilal. He told them that his son would take them to the brothers and wished them luck on their difficult task.

As they reached the brothers abode, they found two identical lodges side by side. Talha couldn't shake off the feeling of anticipation mingled with apprehension. In the left lodge's small garden, they saw a man with strong muscles mending the flower bushes with practiced precision. Bilal saluted him with respect and addressed him as Saad. Saad acknowledged them with a gentle smile, paused in his task, and stood up. Talha could sense Saad's curiosity in his eyes about them. As Bilal was about to introduce them to Saad, they heard a horse's hoof. They all turned towards the noise, and Talha found himself looking at a thin, bearded man with a stern, serious face, but

his eyes were shining with purpose and determination. As the man got down from the horse, Bilal addressed him as Hashir. Talha was astonished to see two very different personalities. As Saad was a tall, gentle-looking, muscular man, Hashir was short in height and rather remote looking. Bilal looked happy for the presence of both brothers and he introduced Moiz and Talha to them. Even though the brothers ignored each other, their attitudes towards them were cordial and welcoming. This encouraged Moiz to cautiously shed light on their visit.

As soon as the brothers heard the request for their collaborative efforts in restoring broken scripture, both became tense, and Talha could sense nerveracking tension in the air. Bilal seemed quiet and wished he could disappear from that place. Moiz continued to appeal to their unique abilities, emphasizing the value of collaboration.

Reimagining the Broken: Saad the Generator

After a while Saad looked intrigued a little bit and seemed open to the idea. However, Hashir, true to his skeptical nature, remained hesitant. With careful and skill handling nature, Moiz tried his best with patience to convey the transformative potential of collaboration, where the strengths of one compensated for the weaknesses of another. How diverse abilities, when harmoniously combined, could yield extraordinary results. At last, Hashir started to show interest and soon both brothers were engaged in discussing ideas for restoration.

Saad told them about his role, that he generated synthetic samples of the given material (random noise). His goal was to make a sample like a real one. Talha associated this with the basic principle of generator in GAN model. First, Saad looked at the scriptures keenly and then began generating the first sample of complete original scripture. It was all fascinating for Talha.

Guidance of History: Hashir the

Discriminator

As the first sample was completed, the look on Saad's face was one of disappointment. Due to a lot of missing text, Saad filled them with random text and images as per his understanding. However, according to him, the combined effort could not be interpreted as meaningful text. Hashir pitched in and examined the resulting work with his discerning abilities, identifying mistakes, such as irrelevant text. After a while, he provided Saad with his critical analysis. Upon listening to Hashir's views, Saad began anew, generating content while avoiding the mistakes pointed out by Hashir. Thus, the process of generation and discrimination continued. Saad produced multiple samples, and after each one underwent careful examination by Hashir's critical eye, he refined his work, avoiding the previous mistakes.

Balancing Act: The Min-Max Game

Saad and Hashir worked on the Min-Max optimization principle^[16], wherein the generator (Saad) aimed to minimize the objective, while the discriminator (Hashir) aimed to maximize the objective. The discriminator's goal was to challenge the generator to improve its generalization accurately. During their collaboration, Hashir explained to Talha the importance of strict discrimination; he emphasized that the stricter the discriminator, the better the quality of Saad's generated output would be. However, this was only achievable if Saad had the courage to tolerate the mistakes pointed out by Hashir. The broken scripture was revived. All of them exchanged triumphant glances as they witnessed the success of their collaboration. Talha and Moiz were ecstatic; not only had they solved the tribe's problem, but they had also brought both brothers together in harmony. They also learned a new concept in the journey of exploring this deep neural network. It's time for them to depart from the tribe and travel to their next intended location. Walking together in the last hours of the night, Moiz and Talha's footfall echoed the beat of a destiny they shared. While walking through the woods, Moiz told

Talha about the several real-world revolutions brought about by the neural network model known as the GAN. For example, GAN facilitates virtual try-on experiences in the fashion business, letting clients picture themselves in various sizes and types of apparel without having to go to a fitting room in person. Before a buyer makes a purchase, GANs can produce pictures of models wearing different outfits, giving them an idea of how the dress would look on them. Talha showed Moiz the illustration of GAN^[17] he made based on the catalogued information gathered from the GAN's tribe as shown in Figure 4.

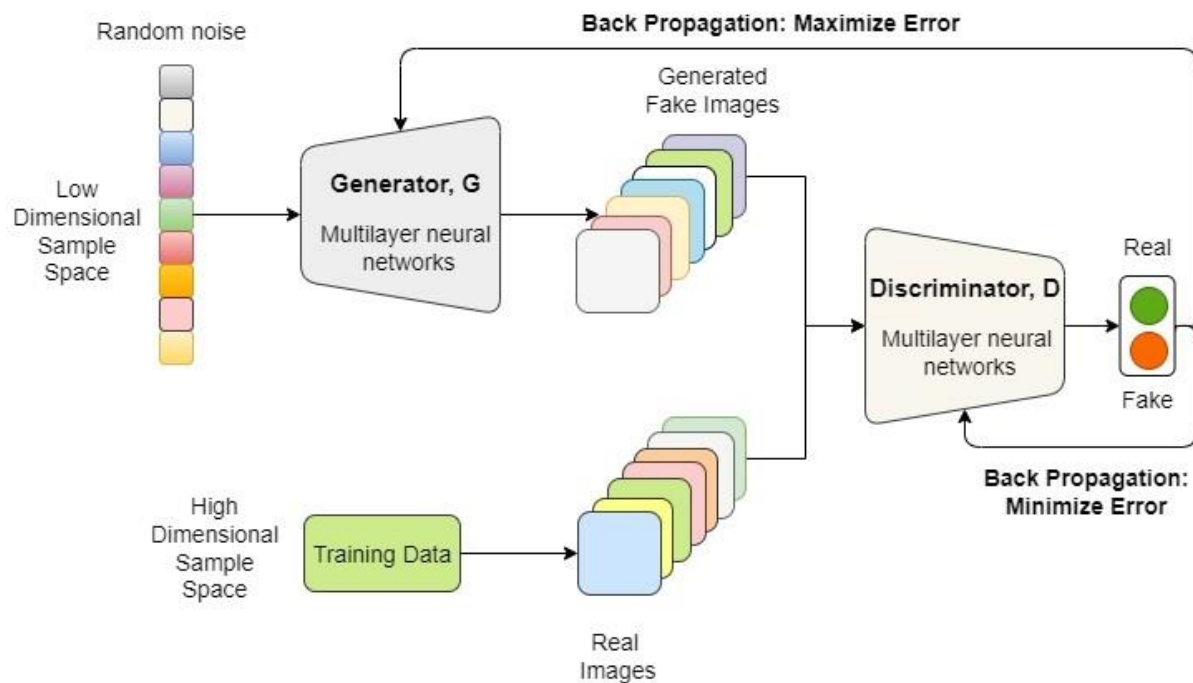


Figure 4: Block Diagram of Generative Adversarial Network

Narrative Arc:

As they commence on their new quest, of getting to know the GAN tribe they observe the dynamic interaction between the discriminator and generator, two adversarial networks. They find themselves in awe of the GAN tribe's capacity to produce lifelike data and visuals, as well as the mechanisms' possible uses in gaming, art, and other fields.



Chapter # 5

Visiting The Transformer Tribe

THE WHISPERING PLAINS OF THE TRANSFORMERS

While crisscrossing towards their final destination according to the map, both were chatting candidly about their feelings regarding their adventure thus far. They were both enthralled by the new knowledge and techniques they had acquired, as well as the power of human skills. They were so engrossed in their conversation that not only did they miss the guidance sign partially obscured by green leaves of a tree, but they also failed to notice that they were being followed. Moiz was the most excited one, and in his excitement, he revealed to Talha that another hidden reason for him to accompany Talha was that he was in possession of an old scripture which he was unable to decipher by himself, as well as by his well-skilled friends and professionals in the field of deciphering. He was hoping that this journey to these tribes might solve this mystery, but yet no tribe had the skill to do so. The GAN tribe had the ability to restore the original text, but still, they did not have the skill to decipher the meaning of the text. However, he was hopeful that this last destination might be helpful in this plight. Talha was astonished to hear about Moiz little secret, and he requested to see it.

As Moiz was showing Talha his prized possession, they were surrounded by five young but sturdy men armed with knives and arrows. Both Talha and Moiz felt their presence when one of them snatched the scripture from Moiz. Those men were looking unfriendly. The one who snatched the relic observed it keenly and then addressed Moiz, asking, "Where did you get it?" Moiz was shocked and therefore immediately could not utter a single word, but when asked again angrily, he hesitantly answered that he found it in one tribe when he visited some years back. The man did not look convinced and consulted with another one. Then he told them that they were captives now and they would be taken to their tribe. Talha was very concerned about their safety. He asked the man about his tribe and why it was necessary for them to go

with them. The man did not answer but just signaled them to follow. When Moiz hesitated to move, the skinny one among the five placed his arrow at the back of Moiz and left no choice for Moiz but to move in the direction mentioned to them. Talha walked behind Moiz in apprehension.

It was mid-afternoon when they reached an opening among the dense trees and saw a number of lodges spread within a vast clear area. It was quiet, as it was traditional siesta time. They were taken inside a heavily guarded lodge and chained to posts in a locked room. Only water was provided to quench their thirst. Talha was really frightened now; even Moiz looked worried. Talha had only heard stories about hostile and unyielding tribes, but up until now, the previous ones had been friendly and welcoming. Not once had he felt insecure. All of their possessions had been taken from them, so they could not even check the time. After an hour or so, a man brought some food. They ate it eagerly, as both of them were hungry. They were then told to move outside and were taken to a large open space where some men were gathered. It appeared they were waiting for them.



In the center of the crowd, one man stood out; he was wearing a gown different from the others. He was a tall, middle-aged man, a bit bulky. He looked at them with an unsmiling face for a while and then started an inquisition about why they were roaming in the jungle and what their purpose was. He asked how they acquired the scripture. Moiz answered all his questions and even told him about their experiences with other tribes in that jungle. When Moiz mentioned GAN's tribe, the man's features softened slightly, but his questions now focused on their time with GAN's tribe. Moiz cautiously told him everything about how they helped GAN restore their holy scripture, which had been lost for years. The man continued to question Moiz about GAN's working method of restoring broken scriptures. Other men were keenly listening to the inquisition with genuine interest. The last question asked was why the scripture in possession with Moiz was not solved

by GAN. Moiz replied that while GAN could restore the broken to the original or create an exact replica of the real, they could not interpret the meaning of the text. They lacked the skill of decoding. Moiz told them that he was specifically looking for a tribe called “Transformers”^[18] who could help him in decoding this scripture as told by one of his professional friends in the real world. Transformer folks were like ancient script whisperers. They were known for cracking the code of ancient scriptures and were adept at all sorts of tasks, like playing with language and understanding visuals. These folks weren't just smart; they could understand an impressive 100 languages, some so old and rare that no one else in the world could read or comprehend them. As Moiz told them more about Transformers, Talha could sense that all of a sudden, the tribe members were now smiling, but with a look that said, “We know more than you”.

The man in the gown chuckled, a deep, rumbling sound. “Moiz,” he said, his voice surprisingly warm, “you've come to the right place. We, the Transformers, are indeed masters of deciphering ancient scripts.” Relief washed over Moiz and Talha. The leader Mehdi, explained their methods. “It's not unlike a captivating story,” he said, his eyes twinkling. “Imagine the scripture as a long, winding journey”.

Moiz and Talha exchanged stunned glances. The man continued, explaining that their tribe, hidden deep within the jungle, had indeed mastered the art of deciphering ancient scripts. They used a complex method that combined ancestral knowledge with a unique understanding of language patterns – a method eerily similar to the workings of the deep neural network variant “Transformers” that Talha had been studying in the university.

Mehdi apologized for their inhospitable attitude in the beginning but attributed it to the nuisance of unfriendly intruders. He then invited them to stay as guests while they examined the scripture. Hope surged through Moiz. Perhaps this was the answer he'd been searching for all along. Mehdi and his most skilled linguists meticulously studied the scripture. They poured over its faded symbols, analyzed its structure, and compared it to fragments of other ancient texts in their possession. Moiz and Talha watched in awe as the

Transformers, with their deep understanding of language, started to unravel the mysteries hidden within the script. Mehdi turned to Moiz and Talha and explained, "We broke down the steps one by one."

Embedding: The Foundation of Understanding

Ayesha, the aged elder with a kind smile, patiently explained the concept of the first step they took. "Imagine a vast collection of books written in an unknown language," she said. "An embedding is like creating a dictionary for that language. We meticulously analyze each symbol, its shape, its position within other symbols, and any historical context we might have. Then, we assign a unique numerical code, to each symbol. This code captures the core meaning of the symbol and allows us to represent the entire script". Moiz added, "This assigning of a unique identifier is like a digital fingerprint used by computers when training on vast amounts of data to be understood." Talha's eyes lit up. "So, it's like creating a common language between the ancient script and the language it's been interpreted into? Like how we used one-hot encoding in the neural network to represent words as numerical vectors?"

Ayesha beamed. "Exactly! This 'common language' becomes the foundation upon which we build our understanding."

Positional Encoding: The Importance of Order

Ali, the elder with the gruff voice, took over next. "Now, Talha," he said, his gaze sharp, "imagine yourself on a journey. Even with a map, if you don't know the order of landmarks, you might get lost, right?" Talha nodded. "The same applies to the script," Ali continued. "The order of the symbols is crucial. A single misplaced symbol can alter the entire meaning. We use a special method passed down through our ancestors to assign a unique value to each symbol based on its position".

Talha chipped in and said, "In the same way, the computer uses a mathematical formula to assign a unique identification not only to

understand the individual symbols but also their relationship within the sequence, just like how the order of landmarks on the map guided us on our journey."

Attention: Unveiling Hidden Connections

Mehdi, the leader, rejoined them, his eyes twinkling with an inner light. "Now, Talha," he said, "we come to the heart of our method –attention^[19]. Imagine you're not just following a path on your journey, but truly experiencing the landscape. You notice how a rushing river might influence the surrounding vegetation, or how a majestic mountain casts long shadows that affect the local wildlife. Similarly, in the script, each symbol has hidden connections with others. Our attention mechanism is like a powerful lens that allows us to see these connections. It analyzes the relationships between symbols, how they influence and depend on each other, to grasp the context and the overall flow of meaning within the script."

Moiz, ever the curious one, leaned forward. "So, the script is like a conversation happening between the symbols themselves?"

Mehdi chuckled. "Not quite a conversation, Moiz," he said. "But the symbols do hold a kind of dialogue. They influence each other, and by understanding these relationships, we can unlock the true meaning the author intended to convey."

Building the Bridge: Encoder and Decoder

Fatima, the young woman enthusiastically, explained the encoder representation. "Think of it like reaching your destination, Talha," she said. "After embedding, positional encoding, and attention, we have a comprehensive understanding of the individual symbols, their order, and their connections. We combine all this information to create a complete picture - the encoder representation. It's akin to constructing a pathway between the ancient script and our contemporary understanding, a path built

upon the basis of our accumulated wisdom". Fatima further explained, "With the encoder representation in place, it's time for the final step: utilizing the encoder group consists of multilingual experts in our tribe. This group gives us the meaning in the ancient language, but most people don't speak those languages anymore. That's where the decoder comes in," she said, his eyes twinkling. "Imagine the encoder representation as a treasure map, Talha. It tells you where the hidden knowledge is buried, but you still need a tool to unearth it. The decoder is that tool, a skilled translator who can bridge the gap between the ancient language and our own". Moiz looked at Talha and said, "Don't you think the encoder decoder's operation is similar to the working of the Autoencoder tribe?" To which Talha replied affirmatively and asked Fatima, "What's the importance of the decoder here? Is its operation similar to the working of the decoder in Autoencoder's tribe?"

Inside the Decoder: Building the New Sequence

Fatima gestured towards a group of young scholars huddled around a table filled with intricate diagrams. "The decoder works in a similar way to the encoder," she explained, "but with a twist". It doesn't analyze the raw symbols directly. Instead, it takes the encoder representation – the compressed understanding of the ancient script – as its starting point." Moiz, ever curious, chimed in, "So, it's like the decoder is looking at a summary of the ancient text?"

Mehdi replied. "Precisely, Moiz! But the decoder doesn't simply duplicate that summary. It uses it as a foundation to build something entirely new – the translated text in our own language."

Attention in the Decoder: Connecting the Dots

"Here's where things get interesting," Fatima said. "Just like the encoder, the decoder utilizes attention. But this time, it pays attention not only to the

internal relationships within the encoder representation, but also back to the original script itself".

Talha furrowed his brow. "So, the decoder checks the original script even though it doesn't understand the individual symbols?"

"Exactly!" Fatima exclaimed. "The decoder uses the encoder representation as a guide, but it constantly refers back to the original symbols, using a special kind of attention, to ensure the translation stays true to the spirit of the ancient text."

The Final Output: Unveiling the Meaning

Mehdi took over again. "Layer by layer, the decoder builds the translated text, one word at a time. It considers the encoder representation, the relationships between those elements, and how they connect back to the original script. This meticulous process allows the decoder to not just translate the words literally, but also capture the nuances, the emotions, and the overall meaning the author intended to convey." After the step-by-step explanation of their working methods, Mehdi gestured for Talha and Moiz to move towards the spread and enjoy supper while the decoding group decoded and finalized the outcome of their efforts. During supper, Mehdi entertained them with tales of their successful endeavors in deciphering multiple ancient scriptures over the years. Even though Moiz was enjoying the supper and the tales, he silently prayed that the team would be able to decipher his scripture. Mehdi encouraged them to rest, but Moiz declined and keenly awaited the results. Talha started to record the step-by-step process of transformer learned from the tribe by also depicting it as an illustration in his notepad (Figure 5).

Finally, Fatima approached them with a beaming smile and invited them to join the gathering where the team was working. Moiz felt his heart race with anticipation, and Talha couldn't contain his excitement either. They eagerly moved towards the congregation with racing hearts. When they reached

them, they saw the elders stood proudly beside a large, unfolded scroll. Hours of meticulous work culminated in this moment. With a grand gesture, Mehdi gestured towards the scroll. "The answer to your quest, Moiz," he boomed. "The meaning of the ancient scripture, unveiled!"

Mehdi invited Moiz to unfold the scroll and read it, as the text was now in simple English. As Moiz began to read, at first his voice wavered with excitement, but then he calmed down and slowly started to read again. The text unveiled an epic poetry, a lost story of a mythological hero who traveled across magical regions, facing formidable foes and difficult obstacles. It discussed bravery, selflessness, and the enduring value of knowledge. They were all enthralled by the story, which connected with them and helped them to bridge the gap between the past and the present, and when Moiz concluded, there was a deep stillness. For Talha, their journey—driven by inquisitiveness and a desire to learn—had revealed the transformational power of knowing, in whatever form it may manifest—in addition to having unveiled the scripture's mysteries.

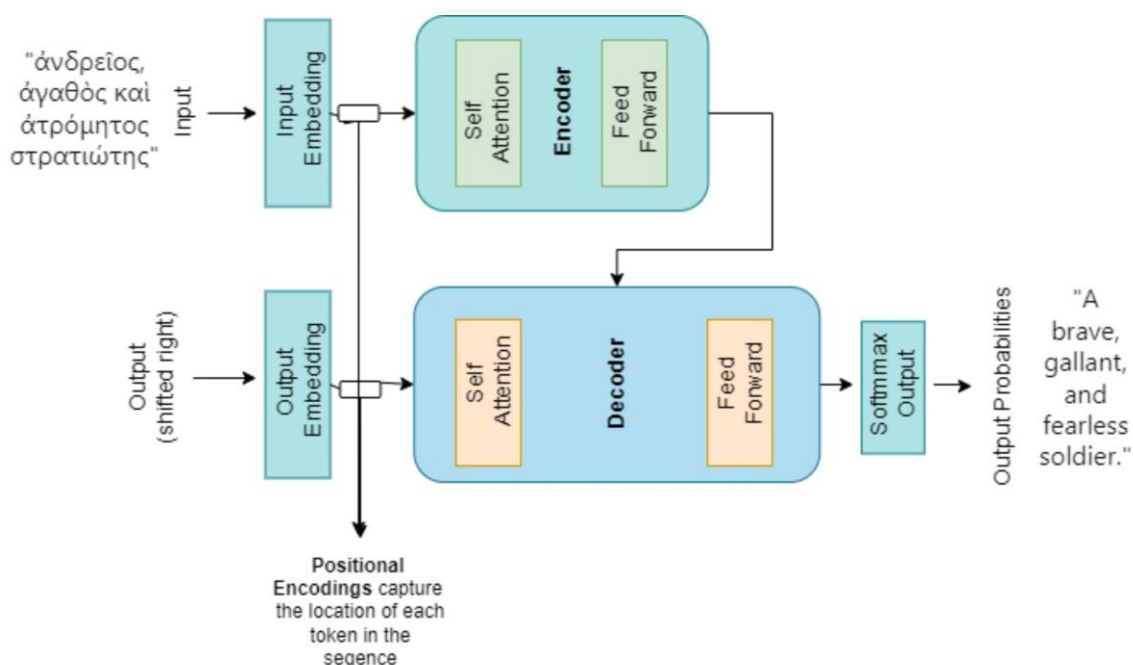


Figure 5: Block Diagram of Transformer Architecture

Explore the Sub-Tribe: Vision Transformer

After exploring the Transformer tribe, Moiz asked the tribe leader, Uzair, "What if we have images and want to extract information from them?" Uzair smiled and led them down a strange path. After walking for some time, they stopped at an entrance gate that looked like it was made of patches. Uzair invited them in. As they walked inside, they saw a whole new world. The people there resembled the Transformer tribe but were well-settled, and the surroundings were very interactive, like painted scenes in images. Both were surprised to see this hidden tribe.

Uzair took them to their tribe leader, Omer. They greeted him, and both Omer and Uzair smiled, seeing the confusion on Talha and Moiz's faces. Without waiting any longer, Moiz asked, "What is this place?" Omer smiled gently and explained, "This tribe is a sub-tribe of the Transformer tribe called the “Vision Transformer”^[20]. We work with images here, just like the Transformers work with text."

Talha and Moiz were amazed. Curiously, Talha asked how it worked. Omer offered them a quick tour of the tribe, which they accepted instantly. As they walked, Omer explained how things worked. They reached a place similar to the Transformer tribe, but here they saw bundles of images—some were crumpled, some very ancient. Omer explained that they extract information from these images, perform classification tasks, and more.

As they continued walking, they saw groups of people making patches of images and passing them as tokens to the next destination. Additional tokens were attached to preserve the spatial and positional information of the images. After that, these tokens were passed to an encoder block, similar to the encoder block of a Transformer, and then to apply the attention mechanism. After encoding, then classification is performed, and important information from the images is preserved.

Omer explained that, in the Transformer tribe, text or sentences are input in the form of tokens. The same concept is applied to images. They make

patches of images with some extra information, like positional information, and input these tokens into the Vision Transformer encoder. Talha drew the block diagram of Vision Transformer (ViT) in notepad capturing the inner working of Transformer (Figure 6).

Omer also told them that there are transformers which uses CNN layers in their architecture to capture both local and global features which are called “Hybrid Transformers”^[21]. He explained to them that the main difference between CNN and Transformer is in the way they capture features. CNN capture local range features and work as a mask but Transformers captures long range dependency features with the help of their attention mechanism.

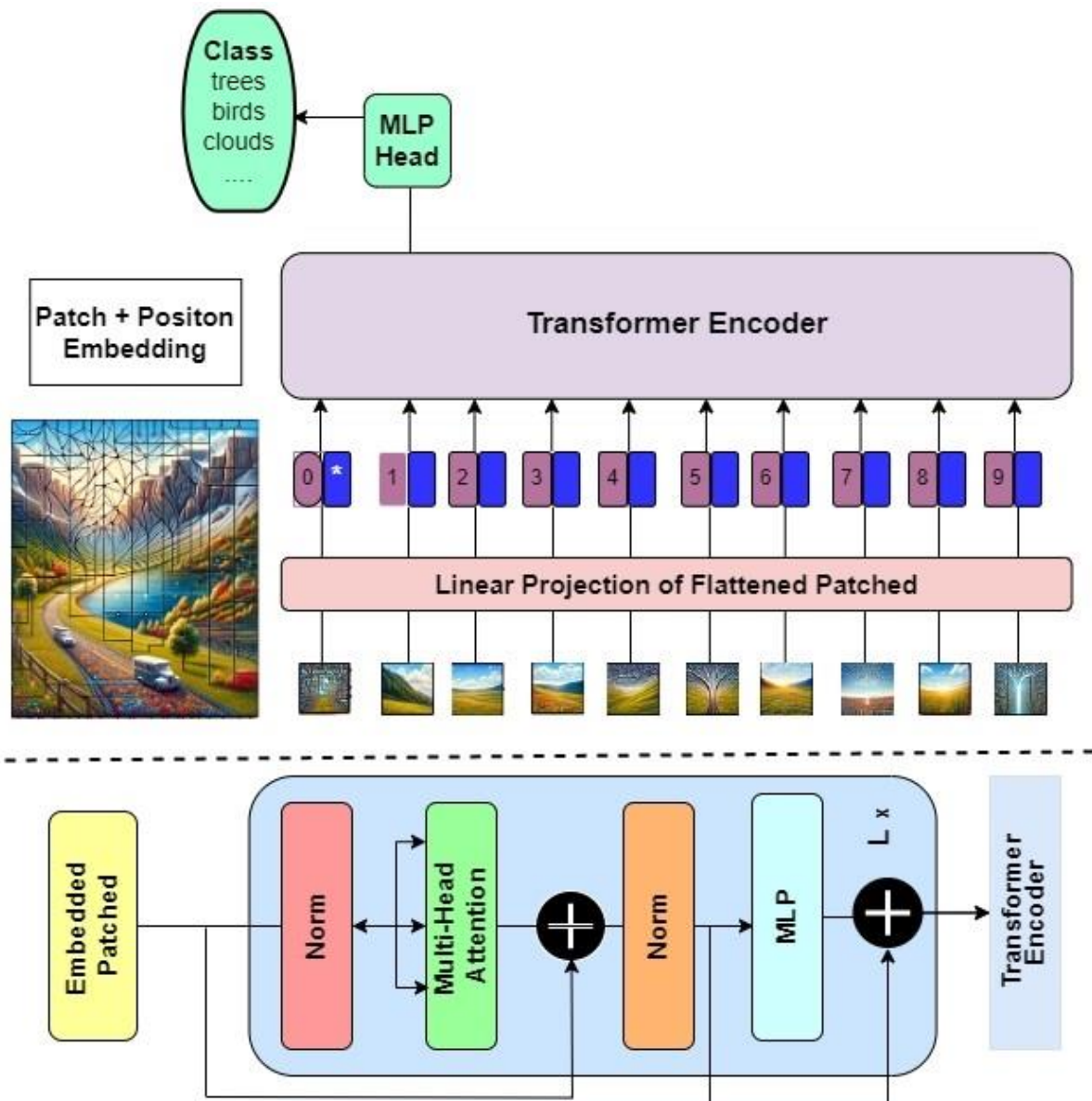
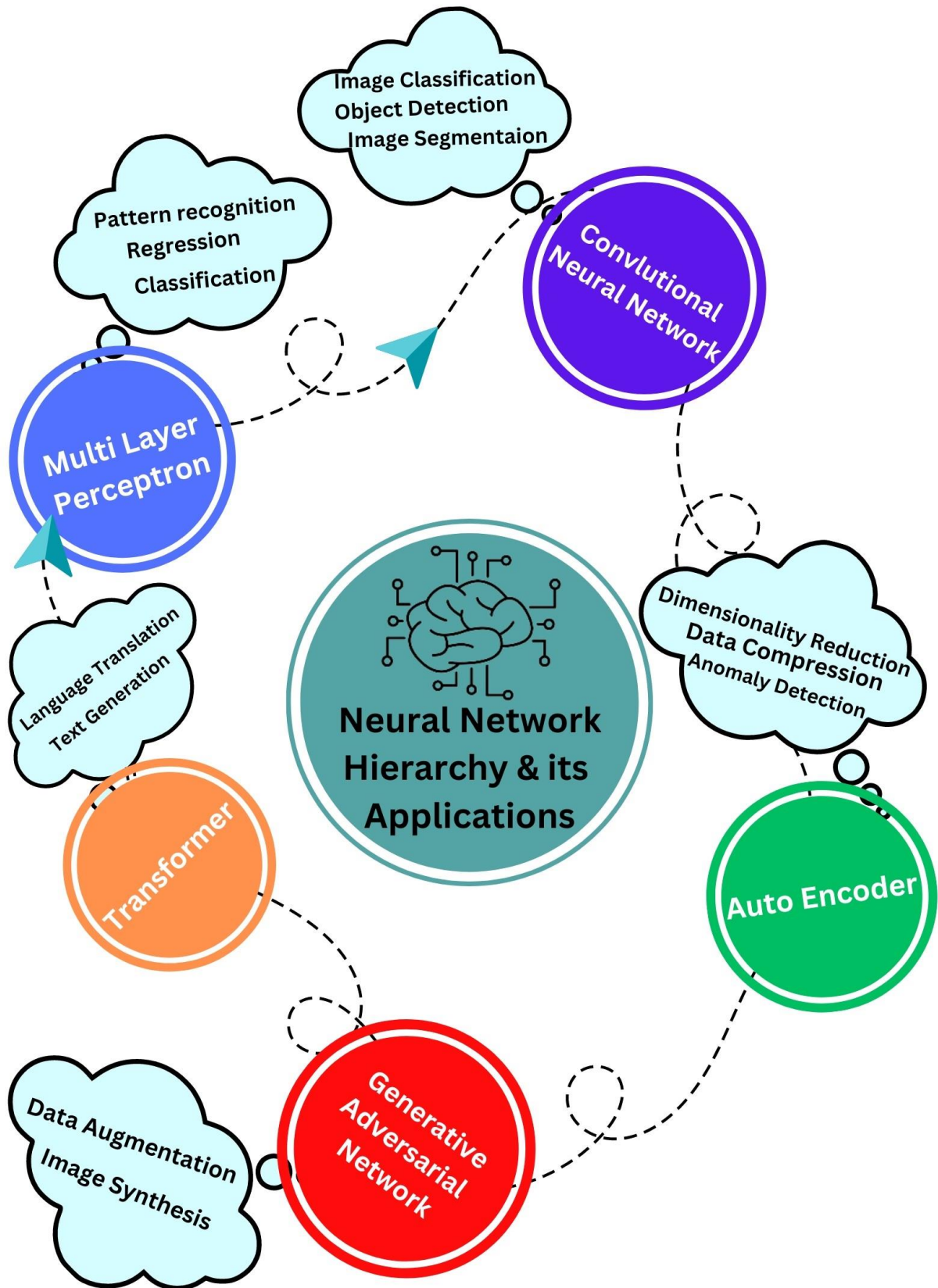


Figure 6: Block Diagram of Vision Transformer Architecture

Narrative Arc:

Ultimately, Talha and Moiz arrive at the culminating point of their adventure: The Transformers tribe. Here, they discover the ground-breaking idea of attention mechanisms—which enable effective input processing and flow in neural networks. They witness how Transformers can revolutionize fields like machine translation and natural language processing. Likewise, they also learn about different architectures of Transformers including vision transformer and Hybrid Transformer.

Following their escape from the neural network wilderness, Talha and Moiz return with a fresh perspective on artificial intelligence and its ability to influence the course of human history. Their journey opens the door to a new era of creativity and exploration by acting as a beacon of information for individuals, who wish to further investigate about the neural networks.



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[1]

“The **multilayer perceptron** is the most known and most frequently used type of neural network. On most occasions, the signals are transmitted within the network in one direction: from input to output. There is no loop, the output of each neuron does not affect the neuron itself. This architecture is called feedforward”[1].

[2]

“**Artificial neural networks (ANNs)** are biologically inspired computer programs designed to simulate the way in which the human brain processes information”[2].

[3]

“**Perceptron** is the most basic model among the various artificial neural nets, has historically impacted and initiated the research in the field of artificial nets, with intrinsic learning algorithm and classification property”[3].

[4]

“ In **classification**, the task is to assign the training input to one of the predefined classes”[4].

[5]

“**Regression** is a mathematical approach used to perform predictive analysis. When the prediction goal is continuous, regression is the appropriate method to use”[4].

[6]

“**Activation Functions** are the function $g(.)$, which is named as activation function, is used to simulate the response state of biological neuron and obtain the output y ”[5].

[7]

“**Back propagation** is a method used in artificial neural networks to calculate a gradient that is needed in the calculation of the weights to be used in the network”[6].

[8]

“**MLP** consists of multiple layers of neurons in a feed-forward architecture. It uses nonlinear activation function with the neurons and each layer is fully connected to the next layer”[7].

[9]

“CNNs are one of the best learning algorithms for understanding image content and have shown exemplary performance in image segmentation, classification, detection, and retrieval related tasks”[8].

[10]

“**Feature Extraction** is a more general method that aims to develop a unique space in which important features of data is stored”[9].

[11]

“The **Pooling Layer** is obtained by applying pooling operator to aggregate information within each small region of the input feature channels and then down sampling the results”[9].

[12]

“**Fully Connected Layer** is a special case of convolutional layer comes up with a method to convert multidimensional convolution operations to 1D convolution operations”[10].

[13]

“The main components of **Autoencoder** are three: an encoder, a latent feature representation, and a decoder”[11]. [14]

The concept of GAN was initially described in 2014 by Goodfellow et al[12] based on “two-player zero-sum game which the total gains of two players are zero, and each player’s gain or loss of utility is exactly balanced by the loss or gain of the utility of another player”[13]. In GANs, there are two neural networks pitted against each other’s as adversaries as Generator and discriminator[12], [13], [14].

[15]

Generator(G)- “tries to capture the potential distribution of real samples, and generates new data samples”[13] i.e. the purpose is to fool the discriminator by generating near-to-realistic samples [14]. Discriminator(D)- “learns to determine whether a sample is from the model distribution(G) or the data distribution”[12] i.e. it tries to discriminate real samples from the generated ones with utmost precision[13].

[16]

This recursive approach is applied in game theory, decision-making, and AI, where a player seeks an optimal move under the assumption that the opponent is also playing optimally. In this scenario, if one player seeks to maximize (MAX) their advantage, the other player seeks to minimize (MIN) it. “Generative model is viewed as a mini-max game consisting of two players (i.e., discriminator and generator)”[15]. [17]

“**Generative Adversarial Network** is one of the most powerful model in unsupervised learning of image generation.”[16].

[18]

“The **Transformer** is the first transduction model relying entirely on self-attention to compute representations of its input and output without using sequence aligned RNNs or convolution”[17].

[19]

“Self-attention estimates the relevance of one item to other items (e.g., which words are likely to come together in a sentence). It is an integral component of Transformers” [18].

[20]

“Vision Transformer (ViT) replaces standard convolutions in deep neural networks on large scale image datasets” [18], [19]. [21]

“Convolutions do an excellent job at capturing low-level local features in images, and have been explored in multiple hybrid ViT designs and specially at the beginning” [18]. “These hybrid vision transformers, also referred to as CNN-Transformer architectures, have demonstrated remarkable results in vision applications” [19].